Status	Finished
Started	Tuesday, 20 October 2026, 4:45 PM
Completed	Tuesday, 20 October 2026, 4:48 PM
Duration	3 mins 22 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Quick Sort
b. Insertion Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **2**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort cannot handle duplicate elements
b. Bubble Sort is difficult to implement
c. Bubble Sort does not guarantee sorted order
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are combined to form a single sorted list
c. They are compared element by element
d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **4**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Sorting the entire list in one pass
- c. Bubbling up the smallest element
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Binary Search
- b. Merge Sort
- c. Linear Search
- d. Bubble Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It builds a sorted array one element at a time
- b. It repeatedly finds the minimum element and moves it to the sorted part of the list
- c. It compares adjacent elements and swaps them if necessary
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **7**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It only works on small datasets
c. It avoids the need for recursion
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **8**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,80,90,150,200,300,400,470,70,120
b. 40,70,80,90,120,150,200,300,400,470
c. 80,150,200,470,40,90,300,400,70,120
d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting decreases the efficiency of selection operations
b. Sorting helps in finding duplicates quickly
c. Sorting makes it harder to search for items
d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Merge Sort
c. Bubble Sort
d. Insertion Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By breaking the input into smaller parts and merging them
b. By comparing elements sequentially
c. By using the bubble-up method
d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **12**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Arranging data in a particular format
c. Searching for data in a list
d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **13**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Bubbling up the largest element to its correct position with each pass
- b. Sorting data in a non-sequential manner
- c. Being the most efficient sorting algorithm
- d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Merge Sort
- c. Quick Sort
- d. Heap Sort

The correct answer is: Bubble Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Backtracking
- c. Dynamic programming
- d. Greedy method

The correct answer is:
Divide and conquer